

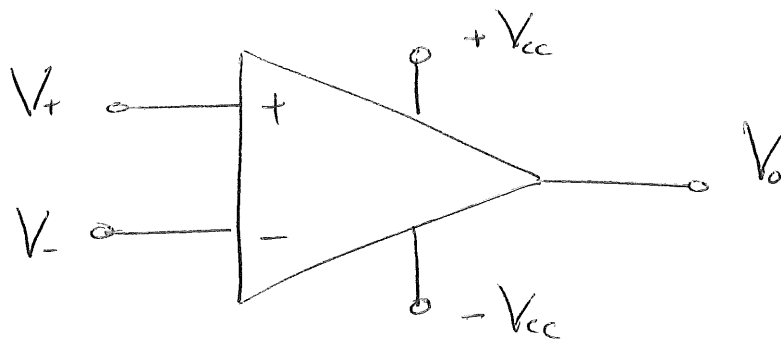
Operasionele Verstekers.

* in Elektroniese eenheid wat optree as 'n
"Voltage Controlled Voltage Source"

* Dit is ontwerp vir wiskundige bewerkinge
(mathematical operations)

$$+ \times - \div \frac{d}{dt} \int$$

→ OpAmp = (Operational Amplifier)



$V_0 \rightarrow$ Uitset

V_+ $\rightarrow$ Nie-omkeers inset

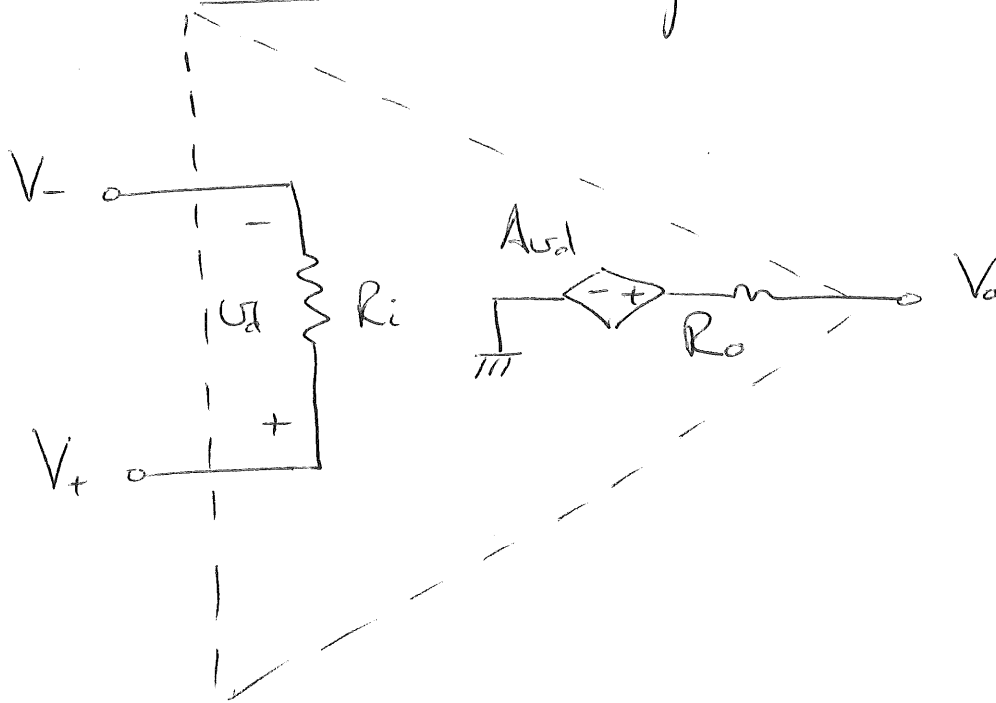
V_- $\rightarrow$ Omkeers inset

$+V_{cc} \rightarrow$ Maksimum limiet vir V_0

$-V_{cc} \rightarrow$ Minimum limiet vir V_0 .

→ Äquivalente Kreislagen

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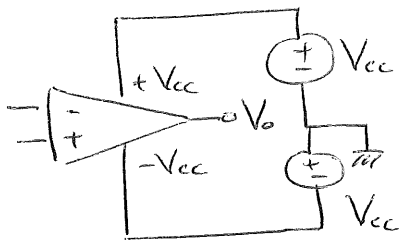


* $u_d = V_+ - V_-$

* $u_o = A u_d$

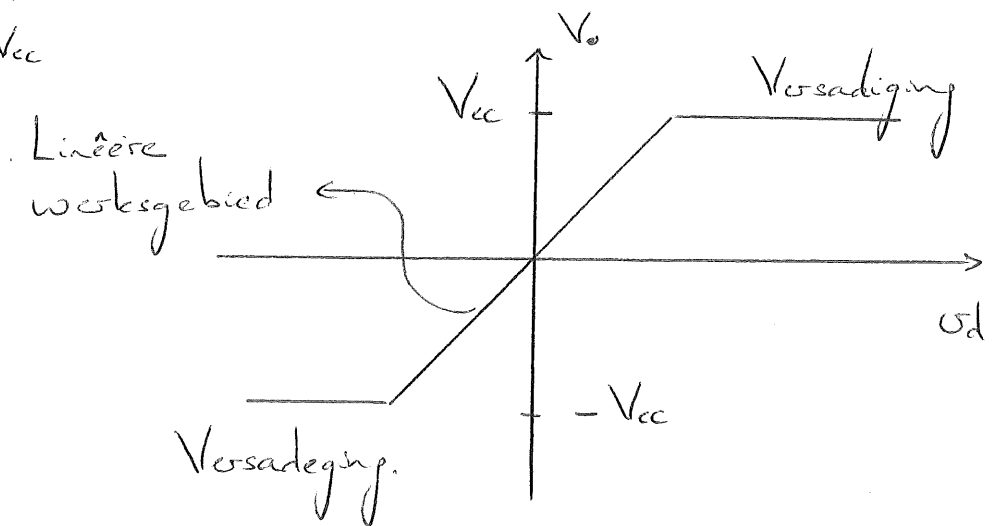
* $A \Rightarrow$ open-loop gain

* Praktische Limit.



* Output is always smaller than $|V_{cc}|$

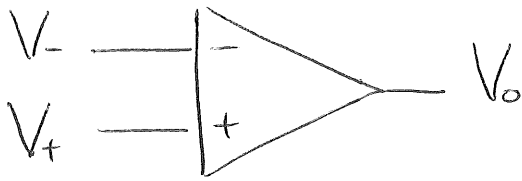
$-V_{cc} < V_o < V_{cc}$



* Ideal vs Non-Ideal

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Ideal



$$A \rightarrow \infty \text{ V/V}$$

$$R_i \rightarrow \infty \Omega$$

$$R_o \rightarrow 0 \Omega$$

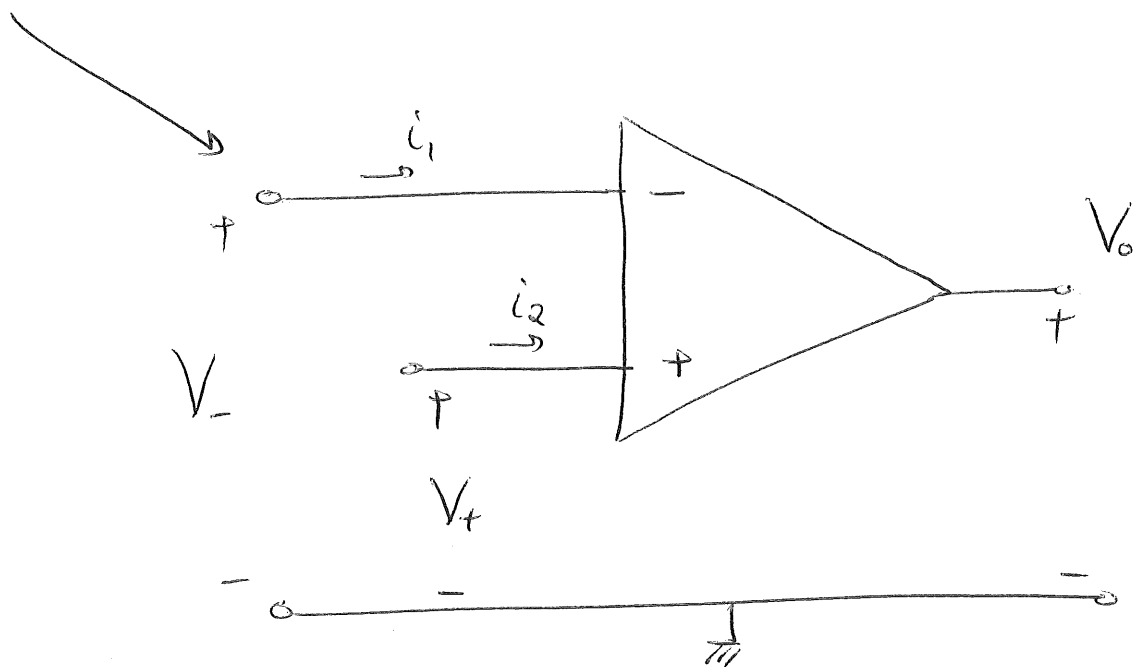
Non-Ideal



$$A = 10^5 - 10^8 \text{ V/V}$$

$$R_i = 10^5 - 10^{13} \Omega$$

$$R_o = 10 - 100 \Omega$$



* For ideal op-amp:

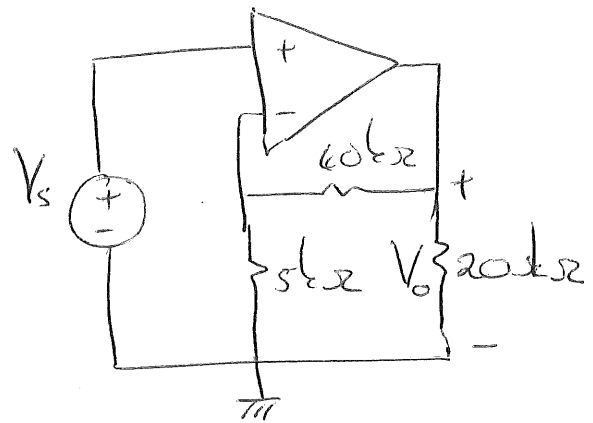
$$i_1 = 0 \text{ A} \quad i_2 = 0 \text{ A}$$

$$V_+ = V_-$$

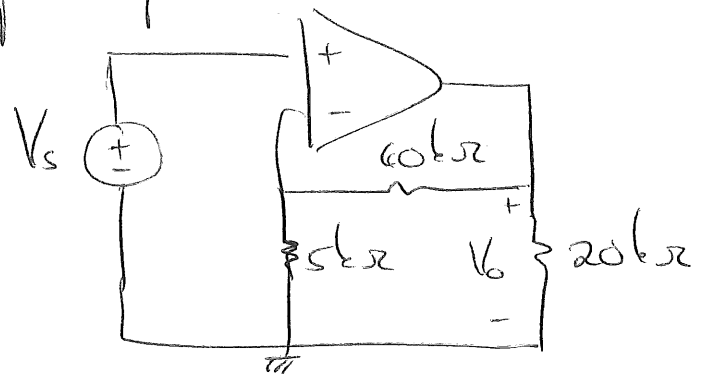
Voorbeeld

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- ① $A = 2 \times 10^5$
 $R_i = 2 \text{ M}\Omega$
 $R_o = 50 \Omega$
 $V_o/V_s = ?$



- ② Aanvaar ideale op-amp.
 $V_o/V_s = ?$



- ③ Ideale op-amps. Bepaal V_o .

